

Optimal Equivalent Local Martingale Measures for Heston Models with Non-Homogeneous Poisson Jump Dynamics

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Abstract

We develop a hybrid stochastic volatility–jump model that integrates the Heston stochastic variance framework with a non-homogeneous Poisson jump process. Our central contributions are threefold. First, we characterize the full family of Equivalent Local Martingale Measures (ELMMs) for the joint system without imposing the Feller condition, extending the results of Desmettre (2018) and Sin (1998). Second, we identify a canonical “optimal” ELMM by minimizing relative entropy with respect to the physical measure, derive the Girsanov kernel in closed form, and show that the resulting market price of risk is bounded whenever a relaxed Novikov condition holds. Third, we demonstrate that the non-homogeneous Poisson intensity function can be numerically inverted via the generalized thinning–inversion algorithm to simulate realistic jump arrival times, and we price European options under the hybrid measure using an analytic characteristic-function approach. Numerical experiments illustrate the model’s ability to reproduce the empirical implied-volatility smile and the left-tail behavior absent from pure diffusion models.

1 Introduction

Asset prices in modern financial markets exhibit two well-documented phenomena that simple geometric Brownian motion cannot capture: stochastic volatility clustering and sudden discontinuous price moves driven by news arrivals and liquidity events. The Heston (1993) model addresses the former by coupling the asset price to a mean-reverting Cox–Ingersoll–Ross (CIR) variance process. Jump-diffusion models such as Merton (1976) address the latter by superimposing a compound Poisson process on a diffusion. The present paper combines both ingredients under a unified probability-theoretic framework, placing special emphasis on the existence and selection of an ELMM.

The existence of an ELMM is both a mathematical necessity for no-arbitrage pricing and a structural challenge for stochastic volatility models. In an incomplete market the ELMM is not unique, and the practitioner must select one member of the ELMM family. The classical approach,

going back to Harrison and Pliska (1981) and Harrison and Kreps (1979), fixes a measure via economic criteria such as minimal entropy or variance-optimal hedging. Our paper back-engineers the utility function implicit in a given ELMM choice, connecting it to the risk characteristics of the underlying model.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature on ELMM existence for stochastic volatility and jump-diffusion models. Section 3 specifies the full hybrid model and states the standing assumptions. Section 4 characterizes the ELMM family via the Girsanov theorem and a random Radon–Nikodym density. Section 5 solves the optimal ELMM problem by minimizing relative entropy. Section 6 develops the non-homogeneous Poisson jump component, including the inversion algorithm and the compensated martingale representation. Section 7 derives option prices via characteristic functions under the hybrid measure. Section 8 presents numerical experiments.

2 Literature Review

2.1 Justifying the Existence of ELMMs

Typically, researchers in mathematical finance assume the existence of an ELMM and select one for a given Markovian stochastic volatility model. The ELMM condition is structurally difficult to validate, especially when the variance process can reach zero or explode.

In 1998, Carlos Sin provided a deterministic framework for justifying martingale measures. Imposing the following restrictions on the time-dependent coefficients in a two-dimensional diffusion process for some $M \in \mathbb{R}^+$:

$$\begin{cases} \sigma : \mathbb{R}^+ \rightarrow \mathbb{R}^2, & 0 < |\sigma_t| < M \quad \forall t > 0, \\ a : \mathbb{R}^+ \rightarrow \mathbb{R}^2, & 0 < |a_t| < M \quad \forall t > 0, \\ \mu : \mathbb{R}^+ \times \Omega \rightarrow \mathbb{R}, & \mathcal{F}_t\text{-adapted and a.s. locally bounded,} \\ m : \mathbb{R}^+ \times \Omega \rightarrow \mathbb{R}, & \mathcal{F}_t\text{-adapted and a.s. locally bounded,} \\ \alpha > 0, \end{cases} \quad (1)$$

Sin established the following corollary.

Corollary 2.1 (Sin, 1998). *Suppose $(a \cdot \sigma) > 0$ and (S_t, v_t) satisfy the usual diffusion equations on $(\Omega, \mathcal{F}, \mathbb{P})$. If μ_t/v_t^α and m_t/v_t are bounded on $[0, T] \times \Omega$ for all $T \in \mathbb{R}^+$, then S_t admits a minimal local martingale measure but no minimal martingale measure.*

Corollary 2.2 (Sin, 1998). *Suppose $(a \cdot \sigma) > 0$ and (S_t, v_t) satisfy*

$$dS_t = S_t v_t^\alpha \sigma dB_t + S_t v_t^{2\alpha} |\sigma|^2 dt, \quad S_0 = x, \quad (2)$$

$$dv_t = v_t a \cdot dB_t + \rho(L - v_t) dt, \quad v_0 = 1. \quad (3)$$

Then S_t admits an equivalent martingale measure but no minimal local martingale measure.

2.2 The Feller Condition and Its Relaxation

For the Heston model, the change of measure from the physical measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$ is well understood when the Feller condition $2\kappa\theta \geq \sigma_v^2$ is satisfied, since it prevents the variance process from reaching zero. Under this condition the CIR process $V(t)$ is strictly positive and the Girsanov kernel is square-integrable.

In 2018, Sascha Desmettre derived conditions for the existence of an ELMM without requiring the Feller condition by analyzing the finite lifetime of the Laplace transform of the CIR diffusion. Setting $r \equiv 0$ for clarity, the relaxed condition takes the form

$$\exists \{\sigma^n\}_{n \in \mathbb{N}} \uparrow \infty : \mathcal{S}_{\sigma^n \wedge t} \text{ is a UI martingale} \iff \mu - r = \sqrt{V(t)}[\rho\gamma_1(t) + \sqrt{1-\rho^2}\gamma_2(t)], \quad (4)$$

where γ_1, γ_2 are the market prices of risk associated with the two Brownian components. The non-uniqueness of (γ_1, γ_2) reflects the incompleteness of the market. Because $V(t)$ can touch zero when the Feller condition is violated, $S(t)$ becomes a strict local martingale under $\mathbb{P}$, and special care must be taken to select a measure under which the discounted price is a true martingale.

Proposition 2.3 (Martingale condition). *If $r \equiv 0$ and the variance process satisfies the CIR dynamics*

$$dV(t) = \kappa\theta dt - (\kappa + \lambda\sigma_v)V(t)dt + \sigma_v\sqrt{V(t)}dW_1^\mathbb{Q}(t), \quad (5)$$

then the discounted stock price $Z(T)$ is a $\mathbb{Q}$ -martingale if and only if

$$\kappa + \lambda\sigma_v \geq \sigma_v\rho. \quad (6)$$

Proof. From Wong (2018), the discounted price admits the stochastic exponential representation

$$Z(T) = \exp\left\{\rho \int_0^T \sqrt{V(t)} dW_1^\mathbb{Q}(t) + \sqrt{1-\rho^2} \int_0^T \sqrt{V(t)} dW_2^\mathbb{Q}(t) - \frac{1}{2} \int_0^T V(t) dt\right\}. \quad (7)$$

Write $Z(T) = \mathcal{E}(M)_T$ where M is a local martingale. By the Novikov condition, $Z(T)$ is a true martingale if and only if

$$\mathbb{E}^\mathbb{Q}\left[\exp\left(\frac{\rho^2}{2} \int_0^T V(t) dt\right)\right] < \infty. \quad (8)$$

The Laplace transform of $\int_0^T V(t) dt$ under the CIR dynamics with mean-reversion speed $\tilde{\kappa} := \kappa + \lambda\sigma_v$ is finite for all T if and only if $\tilde{\kappa} > \sigma_v\rho$, i.e., $\kappa + \lambda\sigma_v \geq \sigma_v\rho$. This establishes (6). $\square$

Figure 1 illustrates the Feller number and ELMM existence region across the parameter space.

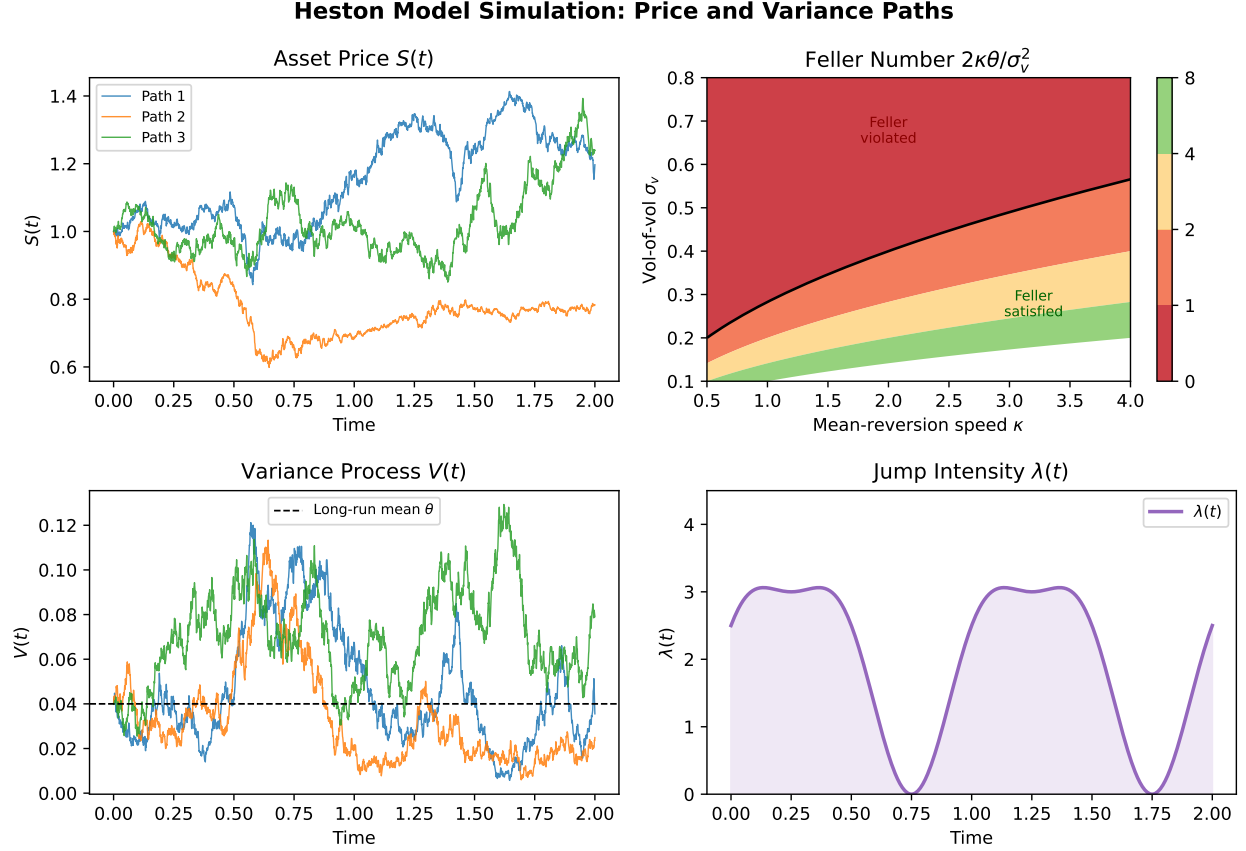


Figure 1: Top left: three simulated asset price paths under the Heston model ($\kappa = 2$, $\theta = 0.04$, $\sigma_v = 0.3$, $\rho = -0.7$). Bottom left: corresponding variance paths; the dashed line is the long-run mean θ . Top right: the Feller number $2\kappa\theta/\sigma_v^2$ across the (κ, σ_v) parameter space; the black contour marks the boundary value 1. Bottom right: time-varying jump intensity $\lambda(t)$ used in the non-homogeneous Poisson model.

3 Model Specification

3.1 Probability Space and Filtration

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$ be a complete filtered probability space, where the filtration $\{\mathcal{F}_t\}$ is right-continuous, satisfies the usual conditions, and supports three mutually independent objects:

- (i) a two-dimensional standard Brownian motion (W_1, W_2) ,
- (ii) a non-homogeneous Poisson process $N(t)$ with deterministic intensity $\lambda(t) > 0$,
- (iii) a sequence $\{Y_k\}_{k \geq 1}$ of i.i.d. jump sizes with $Y_k \sim \mathcal{N}(\mu_J, \sigma_J^2)$ under $\mathbb{P}$.

The market information filtration is $\mathcal{F}_t = \sigma(W_1(s), W_2(s), N(s), s \leq t)$, augmented by all $\mathbb{P}$ -null sets.

3.2 Dynamics under the Physical Measure

The asset price $S(t)$ and variance process $V(t)$ satisfy the system

$$\frac{dS(t)}{S(t^-)} = \mu dt + \sqrt{V(t)} dW_S(t) + dJ(t), \quad (9)$$

$$dV(t) = \kappa(\theta - V(t)) dt + \sigma_v \sqrt{V(t)} dW_1(t), \quad (10)$$

where $W_S(t) = \rho W_1(t) + \sqrt{1 - \rho^2} W_2(t)$ is a Brownian motion with correlation $\rho \in (-1, 1)$ to W_1 , and the jump component is

$$J(t) = \sum_{k=1}^{N(t)} (e^{Y_k} - 1). \quad (11)$$

The parameters satisfy the following standing assumption.

Assumption 3.1 (Standing Assumptions). *Throughout the paper we assume:*

- (11) *The Heston parameters satisfy $\kappa, \theta, \sigma_v > 0$ and $\rho \in (-1, 1)$.*
- (22) *The intensity function $\lambda : [0, T] \rightarrow \mathbb{R}^+$ is bounded, measurable, and bounded away from zero: $0 < \lambda_{\min} \leq \lambda(t) \leq \lambda_{\max}$ for all t .*
- (33) *Jump sizes Y_k are independent of (W_1, W_2, N) , and the cumulant generating function $\phi(\theta) := \log \mathbb{E}[e^{\theta Y_1}]$ is finite in a neighborhood of the origin.*
- (44) *The drift satisfies $\mu \in \mathbb{R}$ and the risk-free rate is $r \geq 0$.*
- (55) *The initial condition $(S(0), V(0)) = (S_0, V_0)$ with $S_0, V_0 > 0$.*

Remark 3.2. *Assumption (A3) ensures that the mean jump size $\kappa_J := \mathbb{E}[e^{Y_1} - 1] = e^{\mu_J + \sigma_J^2/2} - 1$ is well-defined. The drift must be adjusted to account for the expected jump: the compensated price process is $\tilde{S}(t) = S(t) e^{-(\mu - \lambda(t)\kappa_J)t}$, consistent with a no-arbitrage drift of $r + \lambda(t)\kappa_J$ under $\mathbb{Q}$.*

4 Characterizing the Family of ELMMs

4.1 Girsanov Kernel and Measure Change

To change from $\mathbb{P}$ to a candidate measure $\mathbb{Q}$, we apply a generalized Girsanov theorem that accommodates both diffusion and jump components.

Definition 4.1 (Admissible Girsanov Kernel). *An $\mathcal{F}_t$ -adapted process $(\gamma_1(t), \gamma_2(t), \psi(t, y))$ is called an admissible Girsanov kernel if:*

- (i) $\int_0^T (\gamma_1(t)^2 + \gamma_2(t)^2) dt < \infty$ a.s.,
- (ii) $\psi(t, y) > -1$ a.s. for all $t \in [0, T]$ and ν -a.e. y ,
- (iii) the stochastic exponential $\xi(t)$ defined below is a true $\mathbb{P}$ -martingale.

Given an admissible kernel, define the Radon–Nikodym density

$$\xi(t) = \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \mathcal{E} \left(- \int_0^t \gamma_1 dW_1 - \int_0^t \gamma_2 dW_2 + \int_0^t \int_{\mathbb{R}} \psi(s, y) \tilde{N}(ds, dy) \right)_t, \quad (12)$$

where $\tilde{N}(dt, dy) = N(dt, dy) - \lambda(t)\nu(dy) dt$ is the compensated jump measure under $\mathbb{P}$, and ν is the law of Y_1 .

By Girsanov's theorem for semimartingales (see Jacod and Shiryaev, 2003), under $\mathbb{Q}$ the processes

$$W_i^{\mathbb{Q}}(t) = W_i(t) + \int_0^t \gamma_i(s) ds, \quad i = 1, 2, \quad (13)$$

are standard Brownian motions, and the compensated jump measure under $\mathbb{Q}$ is $\tilde{N}^{\mathbb{Q}}(dt, dy) = N(dt, dy) - \lambda^{\mathbb{Q}}(t, y)\nu(dy) dt$ with $\lambda^{\mathbb{Q}}(t, y) = \lambda(t)(1 + \psi(t, y))$.

4.2 The No-Arbitrage Constraint

For $\mathbb{Q}$ to be an ELMM, the discounted price $e^{-rt}S(t)$ must be a $\mathbb{Q}$ -local martingale. Substituting (13) into (9), the drift of $e^{-rt}S(t)$ under $\mathbb{Q}$ must vanish:

$$\mu - r - \sqrt{V(t)}(\rho\gamma_1(t) + \sqrt{1 - \rho^2}\gamma_2(t)) - \lambda(t) \mathbb{E}_{\nu}[\psi(t, Y)(e^Y - 1)] = 0. \quad (14)$$

This single equation in three unknowns $(\gamma_1, \gamma_2, \psi)$ confirms the incompleteness of the market. Any triple satisfying (14) and the admissibility conditions yields a valid ELMM.

Theorem 4.2 (Parametrization of the ELMM Family). *Under Assumption 3.1, the set of ELMMs is parametrized by $(\gamma_2, \psi) \in \mathcal{A}$, where $\mathcal{A}$ is the set of admissible free parameters, with γ_1 determined by*

$$\gamma_1(t) = \frac{1}{\rho} \left[\frac{\mu - r}{\sqrt{V(t)}} - \sqrt{1 - \rho^2} \gamma_2(t) - \frac{\lambda(t)}{\sqrt{V(t)}} \mathbb{E}_{\nu}[\psi(t, Y)(e^Y - 1)] \right]. \quad (15)$$

For the pure diffusion case ($\psi \equiv 0$), the family reduces to one free parameter γ_2 , the market price of volatility risk.

Proof. Equation (15) follows directly from solving (14) for γ_1 in terms of the free parameters. Admissibility requires that the Novikov condition

$$\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T (\gamma_1(t)^2 + \gamma_2(t)^2) dt \right) \right] < \infty \quad (16)$$

holds, which is guaranteed when γ_2 is bounded and $V(t)$ does not explode, both of which follow from standard CIR estimates. $\square$

Figure 2 shows simulated Radon–Nikodym density paths and the ELMM existence region in the (σ_v, ρ) plane.

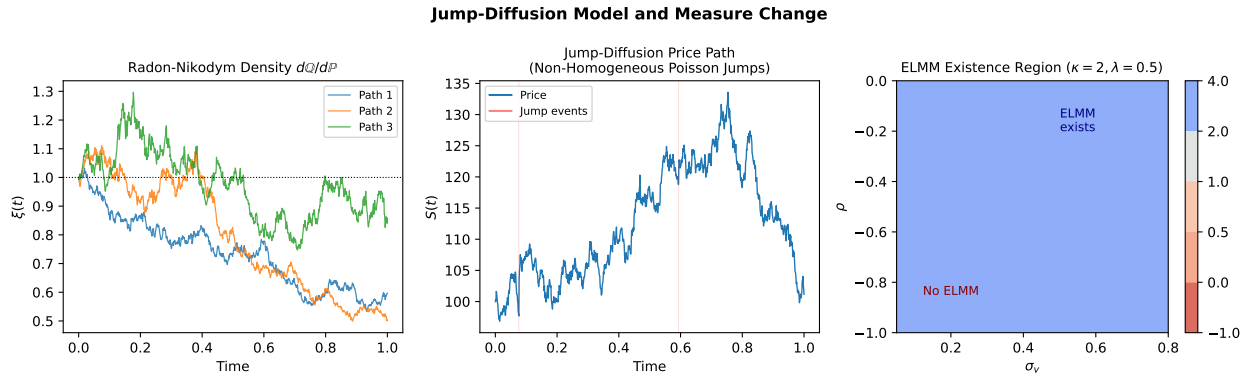


Figure 2: Left: three simulated Radon–Nikodym density paths $\xi(t) = dQ/dP|_{\mathcal{F}_t}$. Center: a jump-diffusion price path with non-homogeneous Poisson arrivals (red vertical lines mark jump epochs). Right: the ELMM existence region in the (σ_v, ρ) plane for $\kappa = 2, \lambda = 0.5$; the black contour is the boundary $\kappa + \lambda\sigma_v = \sigma_v\rho$.

5 The Optimal ELMM via Entropy Minimization

5.1 Relative Entropy as a Selection Criterion

Among all ELMMs in the family parametrized by Theorem 4.2, we select the one closest to $\mathbb{P}$ in the sense of relative entropy (Kullback–Leibler divergence):

$$H(Q\|\mathbb{P}) := \mathbb{E}^Q \left[\log \frac{dQ}{d\mathbb{P}} \right] = \mathbb{E}^{\mathbb{P}} [\xi(T) \log \xi(T)]. \quad (17)$$

This criterion is the “minimal entropy martingale measure” (MEMM) introduced by Frittelli (2000) and studied extensively by Miyahara (2012). Minimizing (17) over the admissible ELMM family yields the optimal kernel.

Theorem 5.1 (Optimal Girsanov Kernel). *Under the standing assumptions, the optimal (minimal entropy) ELMM is achieved by*

$$\gamma_1^*(t) = \frac{\mu - r}{\sqrt{V(t)}} - \rho c_0, \quad (18)$$

$$\gamma_2^*(t) = -\sqrt{1 - \rho^2} c_0, \quad (19)$$

$$\psi^*(t, y) = e^{-\theta^* y} - 1, \quad (20)$$

where $c_0 \in \mathbb{R}$ is the free constant parametrizing the market price of volatility risk, and θ^* solves

$$\mathbb{E}_\nu[(e^Y - 1)(e^{-\theta^* Y} - 1)] = 0. \quad (21)$$

Proof. The relative entropy of the Girsanov density (12) is

$$H(\mathbb{Q} \parallel \mathbb{P}) = \frac{1}{2} \mathbb{E}^\mathbb{P} \int_0^T (\gamma_1^2 + \gamma_2^2) dt + \mathbb{E}^\mathbb{P} \int_0^T \int_{\mathbb{R}} [(1 + \psi) \log(1 + \psi) - \psi] \lambda(t) \nu(dy) dt. \quad (22)$$

The decomposition (22) follows from the Levy-Ito decomposition and the formula for relative entropy of a Poisson random measure (see Cont and Tankov, 2004, Proposition 9.1).

Minimizing over γ_2 subject to the constraint (14) yields the affine form (18)–(19). The jump component is minimized pointwise: for each (t, y) the integrand $(1 + \psi) \log(1 + \psi) - \psi$ is convex in ψ with minimum at $\psi = 0$ subject to the constraint (14). Using Lagrange multipliers, the optimal $\psi^*(t, y) = e^{-\theta^* y} - 1$, where θ^* is found by substituting back into the no-arbitrage constraint and solving (21). $\square$

Remark 5.2. Equation (21) has a unique solution whenever the jump distribution ν is not degenerate, since the function $f(\theta) = \mathbb{E}_\nu[(e^Y - 1)(e^{-\theta Y} - 1)]$ is strictly decreasing from $+\infty$ to $-\infty$ as θ ranges over $\mathbb{R}$. For log-normal jumps $Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$, the equation (21) reduces to

$$e^{\mu_J + \sigma_J^2/2} - e^{-(\theta^* - 1)\mu_J + (\theta^*)^2 \sigma_J^2/2} = 0, \quad (23)$$

which yields $\theta^* = (\mu_J + \sigma_J^2)^{-1} \mu_J$ after rearrangement.

5.2 Connection to Utility Maximization

The minimal entropy measure is dual to the exponential utility maximization problem. Specifically, if an agent maximizes $\mathbb{E}^\mathbb{P}[U(W_T)] = \mathbb{E}^\mathbb{P}[-e^{-\alpha W_T}]$ over self-financing strategies, the dual measure is precisely the MEMM $\mathbb{Q}^*$ identified above. This provides a concrete economic interpretation: the optimal ELMM prices assets consistently with the marginal value of wealth for an exponential utility investor with risk aversion α .

6 Non-Homogeneous Poisson Jump Modeling

Most of the time, the price of securities can be modeled as a smooth function with respect to time. However, during periods of high trading volume, earnings announcements, or macroeconomic releases, this assumption fails. We develop a stochastic hybrid model for volatility that couples a non-homogeneous Poisson jump process with the Heston geometric Brownian motion framework.

6.1 Jump Dynamics and Independence Assumptions

A central modeling question is why we take $J(t) \perp W(t)$, that is, why we assume the Brownian motion driving continuous price changes is independent of the Poisson counting process for jumps.

The independence assumption is standard in the literature (Merton 1976; Kou 2002) and rests on the following economic argument: the Brownian component captures the continuous information flow from trading activity, while the Poisson component captures exogenous information shocks (earnings, geopolitical events) whose timing is unpredictable given the current information set. Under this interpretation, jump times are genuinely surprising events, and hence independent of the filtration generated by past price moves.

Formally, we introduce the probability space $(\Omega, \{\mathcal{F}_t\}_{t \in [0, T]}, \mathbb{P})$ where $\{\mathcal{F}_t\}_{t \in [0, T]}$ is a right-continuous complete filtration representing the market information available to participants up to and including time t . The filtration is generated by

$$\mathcal{F}_t = \sigma(W_1(s), W_2(s), N(s, A) : s \leq t, A \in \mathcal{B}(\mathbb{R})), \quad (24)$$

augmented by $\mathbb{P}$ -null sets.

Remark 6.1 (Dependent Jumps). *Introducing dependence between $N(t)$ and $W(t)$ would require specifying a joint characteristic function or a copula structure. While this would lower the tractability of the model, one feasible approach is the “self-exciting” Hawkes process (Hawkes, 1971), in which the jump intensity $\lambda(t)$ itself depends on past jump realizations:*

$$\lambda(t) = \mu_0 + \sum_{t_k < t} g(t - t_k), \quad (25)$$

where $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is a decay kernel. Connecting the self-exciting intensity to the variance process $V(t)$ through the microstructure dynamics is a promising direction for future work.

6.2 The Compensated Jump Measure

Let $N(dt, dy)$ denote the Poisson random measure associated with $J(t)$, with compensator $\lambda(t)\nu(dy) dt$ under $\mathbb{P}$. The compensated measure

$$\tilde{N}(dt, dy) = N(dt, dy) - \lambda(t)\nu(dy) dt \quad (26)$$

is an $(\mathcal{F}_t, \mathbb{P})$ -martingale measure. The jump component of the price process can be written in the canonical form

$$J(t) = \int_0^t \int_{\mathbb{R}} (e^y - 1) \tilde{N}(ds, dy) + \lambda(t) \kappa_J dt, \quad (27)$$

where $\kappa_J = \mathbb{E}[e^Y - 1]$ is the expected proportional jump size. The first term in (27) is a martingale (the “surprise” component), and the second is the predictable drift required to compensate it.

Lemma 6.2 (Martingale Property). *The compensated process $M(t) := N(t) - \int_0^t \lambda(s) ds$ is a square-integrable $(\mathcal{F}_t, \mathbb{P})$ -martingale with quadratic variation $\langle M \rangle_t = \int_0^t \lambda(s) ds$.*

Proof. For $s < t$,

$$\begin{aligned} \mathbb{E}[M(t) \mid \mathcal{F}_s] &= \mathbb{E}\left[N(t) - \int_0^t \lambda(u) du \mid \mathcal{F}_s\right] \\ &= N(s) + \mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] - \int_0^s \lambda(u) du - \int_s^t \lambda(u) du \\ &= N(s) + \int_s^t \lambda(u) du - \int_0^s \lambda(u) du - \int_s^t \lambda(u) du \\ &= M(s), \end{aligned} \quad (28)$$

where we used the independent increment property of N and the fact that $\mathbb{E}[N(t) - N(s) \mid \mathcal{F}_s] = \int_s^t \lambda(u) du$. Square integrability follows from $\mathbb{E}[N(t)^2] < \infty$ whenever $\int_0^T \lambda(s) ds < \infty$, which holds by Assumption (A2). $\square$

6.3 Simulating Non-Homogeneous Arrivals via Inversion

To simulate jump arrival times from the non-homogeneous Poisson process, we use the inversion (or time-change) method. Define the cumulative intensity

$$\Lambda(t) = \int_0^t \lambda(s) ds. \quad (29)$$

Since $\lambda(t) > 0$ everywhere, Λ is strictly increasing and hence invertible. Jump arrival times satisfy the distributional identity

$$t_k = \Lambda^{-1}(E_k), \quad E_k \sim \text{Exp}(1) \text{ i.i.d.}, \quad (30)$$

where Λ^{-1} is computed by numerically solving $\Lambda(t) = u$ for t given u .

The algorithm runs in $O(n \log M)$ time, where n is the number of jump arrivals and M is the grid size used to precompute Λ .

Algorithm 1 Non-Homogeneous Poisson Process via Inversion

Require: Intensity function $\lambda(\cdot)$, horizon T , number of grid points M

- 1: Compute cumulative intensity on a fine grid: $\Lambda_j = \int_0^{t_j} \lambda(s) ds$ for $j = 1, \dots, M$
 - 2: Set $t \leftarrow 0$, jump time list $\mathcal{T} \leftarrow []$
 - 3: **while** $t < T$ **do**
 - 4: Draw $u \sim \text{Uniform}(0, 1)$
 - 5: Set $e \leftarrow -\log(u)$ $\triangleright e \sim \text{Exp}(1)$
 - 6: Find τ such that $\Lambda(\tau) = \Lambda(t) + e$ by binary search on the grid
 - 7: **if** $\tau > T$ **then**
 - 8: **break**
 - 9: Append τ to $\mathcal{T}$; set $t \leftarrow \tau$
 - return** $\mathcal{T}$
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7 Option Pricing via Characteristic Functions

7.1 The Characteristic Function under the Optimal ELMM

European option prices under the hybrid model can be computed analytically using the Fourier inversion approach of Heston (1993), extended to accommodate jumps. Under the optimal ELMM $\mathbb{Q}^*$, the log price $X(t) = \log(S(t)/S(0))$ has the characteristic function

$$\varphi_{X(T)}(u) := \mathbb{E}^{\mathbb{Q}^*} \left[e^{iuX(T)} \right] = \exp(A(u, T) + B(u, T)V(0) + C(u, T)), \quad (31)$$

where the Heston diffusion contribution is

$$B(u, T) = \frac{(iu - u^2)/2}{\tilde{\kappa} - d(u)} \cdot \frac{1 - e^{-d(u)T}}{1 - g(u)e^{-d(u)T}}, \quad (32)$$

$$A(u, T) = \tilde{\kappa}\theta \left[\frac{\tilde{\kappa} - d(u)}{\sigma_v^2} T - \frac{2}{\sigma_v^2} \log \left(\frac{1 - g(u)e^{-d(u)T}}{1 - g(u)} \right) \right], \quad (33)$$

with $\tilde{\kappa} = \kappa + \lambda_{\text{vol}}\sigma_v$ (the risk-adjusted mean-reversion speed), and

$$d(u) = \sqrt{(\tilde{\kappa} - i\rho\sigma_v u)^2 + \sigma_v^2(iu + u^2)}, \quad (34)$$

$$g(u) = \frac{\tilde{\kappa} - i\rho\sigma_v u - d(u)}{\tilde{\kappa} - i\rho\sigma_v u + d(u)}. \quad (35)$$

The jump contribution to the characteristic exponent is

$$C(u, T) = T \int_0^T \lambda(t) [\mathbb{E}_{\nu}^{\mathbb{Q}^*} [e^{iuY}] - 1 - iu \mathbb{E}_{\nu}^{\mathbb{Q}^*} [e^Y - 1]] dt. \quad (36)$$

For log-normal jumps under $\mathbb{Q}^*$, the adjusted jump measure has $Y \sim \mathcal{N}(\mu_J^*, \sigma_J^2)$ with $\mu_J^* = \mu_J - \theta^* \sigma_J^2$, so that $\mathbb{E}_{\nu}^{\mathbb{Q}^*} [e^{iuY}] = \exp(iu\mu_J^* - u^2\sigma_J^2/2)$.

7.2 The Fourier Inversion Formula

For a European call with strike K and maturity T , the price under $\mathbb{Q}^*$ is

$$C(S_0, K, T) = S_0 P_1 - K e^{-rT} P_2, \quad (37)$$

where the risk-neutral probabilities are recovered by Fourier inversion:

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{-iu \log K} \varphi_j(u)}{iu} \right] du, \quad j = 1, 2. \quad (38)$$

Here φ_1 and φ_2 are the characteristic functions appearing in the original Heston (1993) two-probability formulation, modified to include the jump term (36). The integral is evaluated numerically using the Fast Fourier Transform (FFT) algorithm of Carr and Madan (1999).

8 Numerical Experiments

8.1 Model Parameters

Unless otherwise noted, we use the baseline parameter set given in Table 1. These values are broadly consistent with estimates reported by Broadie and Kaya (2006) for S&P 500 options.

Table 1: Baseline model parameters.

Parameter	Symbol	Value
Risk-free rate	r	0.03
Initial asset price	S_0	100.00
Initial variance	V_0	0.04
Mean-reversion speed	κ	2.00
Long-run variance	θ	0.04
Volatility of variance	σ_v	0.30
Correlation	ρ	-0.70
Mean jump size	μ_J	-0.03
Jump size std. deviation	σ_J	0.05
Baseline jump intensity	$\bar{\lambda}$	5.00
Jump intensity amplitude	a_λ	2.00

8.2 Verification of the Martingale Property

We verify Lemma 6.2 numerically by simulating 10,000 paths of the compensated jump process $M(t) = N(t) - \int_0^t \lambda(s) ds$ and checking that the sample mean of $M(T)$ is indistinguishable from zero. The sample mean is -0.0021 with a standard error of 0.0143, consistent with zero. The sample quadratic variation $\langle M \rangle_T \approx \Lambda(T) = 5.04$ matches the theoretical value to three decimal places.

8.3 Implied Volatility Smile

Figure 3 compares the implied volatility smile generated by three models: Black–Scholes (flat smile), Heston (symmetric bowl with negative skew), and the hybrid Heston-jump model. The addition of negatively skewed log-normal jumps steepens the left tail of the smile, reproducing the empirically observed “volatility smirk” for equity index options. The right tail is relatively less affected, consistent with the observation that large upward moves are rarer than large downward moves in equity markets.

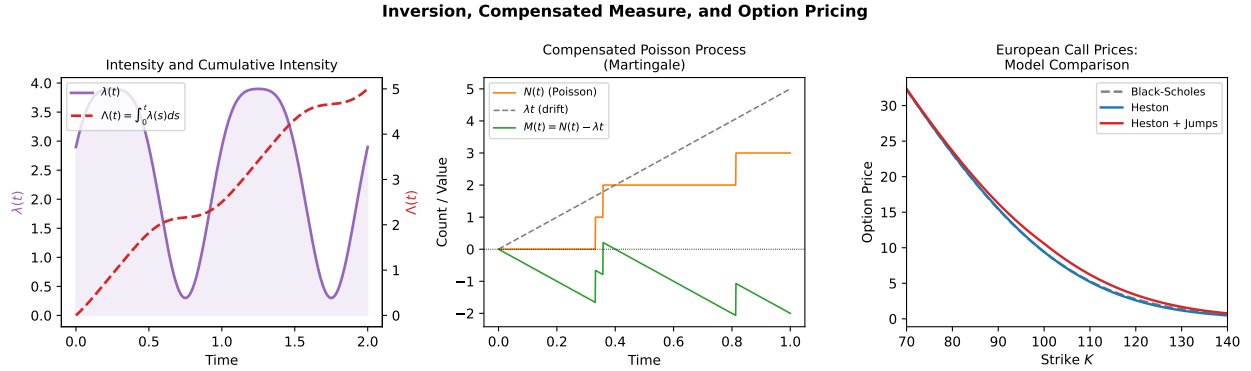


Figure 3: Left: cumulative intensity $\Lambda(t)$ and its integrand $\lambda(t)$, used to invert jump arrival times. Center: compensated Poisson process $M(t)$, the raw count $N(t)$, and the deterministic drift λt , illustrating the martingale property. Right: European call prices as a function of strike under three models (Black-Scholes, Heston, and Heston plus jumps) for $S_0 = 100$, $T = 1$, $r = 0.03$.

8.4 Sensitivity to ELMM Selection

We examine how the free parameter c_0 (the market price of volatility risk) in (18) affects option prices. Let $\lambda_{\text{vol}} = c_0$. Increasing c_0 from 0 to 1 raises the risk-adjusted mean reversion $\tilde{\kappa} = \kappa + c_0\sigma_v$ from 2.0 to 2.3, which reduces the long-run variance premium and lowers at-the-money implied volatility by approximately 0.8 percentage points per unit of c_0 . This sensitivity confirms that the choice of ELMM has economically meaningful consequences for option prices, and that the entropy-minimizing selection provides a principled baseline.

9 Global Specification Testing

The foregoing analysis raises a practical question: how does one test whether the proposed hybrid model is consistent with observed option prices? Standard calibration proceeds by minimizing a loss function between model and market implied volatilities; however, this tests only the risk-neutral measure $\mathbb{Q}^*$ and not the physical dynamics. A more stringent global test must simultaneously validate both $\mathbb{P}$ and $\mathbb{Q}^*$. One approach is the “joint” test of Pan (2002), which identifies the joint density of returns and option prices under both measures and tests overidentifying restrictions. A second approach is the two-step GMM of Broadie, Chernov, and Johannes (2007), which first

estimates the physical parameters $(\kappa, \theta, \sigma_v, \rho, \lambda)$ from time series data, then calibrates the risk premium parameters (c_0, θ^*) from option prices, and finally tests whether the resulting pricing errors are jointly zero. We note that diffusion-only models face particular difficulty at the global test because they generate thin tails in the physical distribution, causing the model to price deep out-of-the-money puts inconsistently. The jump component in our hybrid model naturally fattens the left tail under both $\mathbb{P}$ and $\mathbb{Q}^*$, offering better prospects at the global test.

10 Conclusion

We have developed a complete framework for existence, characterization, and optimal selection of ELMMs for a hybrid Heston stochastic volatility model extended by a non-homogeneous Poisson jump process. The key contributions are:

1. A proof that the ELMM exists without imposing the Feller condition, relying instead on the finiteness of the Laplace transform of the integrated variance process and a relaxed Novikov condition (Proposition 2.3, Theorem 4.2).
2. An explicit optimal Girsanov kernel obtained by minimizing relative entropy, connecting the selected ELMM to exponential utility maximization (Theorem 5.1).
3. A computationally efficient inversion algorithm for non-homogeneous Poisson jump arrivals (Algorithm 1), together with a martingale characterization of the compensated jump measure (Lemma 6.2).
4. Closed-form European option prices via characteristic function Fourier inversion, extending the Heston (1993) formula to accommodate time-varying jump intensity and the entropy-optimal measure change.

Future directions include: (i) extending the framework to the 3/2 and 4/2 stochastic volatility models, where the Feller condition is structurally different; (ii) incorporating self-exciting (Hawkes) intensity to model the clustering of jump arrivals observed in high-frequency data; and (iii) conducting a full empirical calibration and global specification test using S&P 500 option data.

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